

BAHI

Member-Witnessed Verification Layer for the SHG Digital Ledger

Title

The person who records the money is no longer the only proof.

Track

PS-17 SHG Digital Ledger - Tamper-Evident Financial Logs

Team Name.(as registered)

Team 511 (confirm against portal at login)

Problem Statement

1 crore+ SHGs still run on handwritten registers.

Bookkeeping errors, disputes over who owes what, and embezzlement by office-bearers often go undetected until the savings are gone.

Scale: 10.03 crore women members (DAY-NRLM), 144.22 lakh savings-linked SHG bank accounts (NABARD).

LokOS already digitizes the books (94.16 lakh SHGs). But it is server-authoritative: a member cannot verify a single entry about her own money, offline, ever.

The gap is not digitization. It is verification.

Technical Approach

Stack and method, chosen for the 7h window

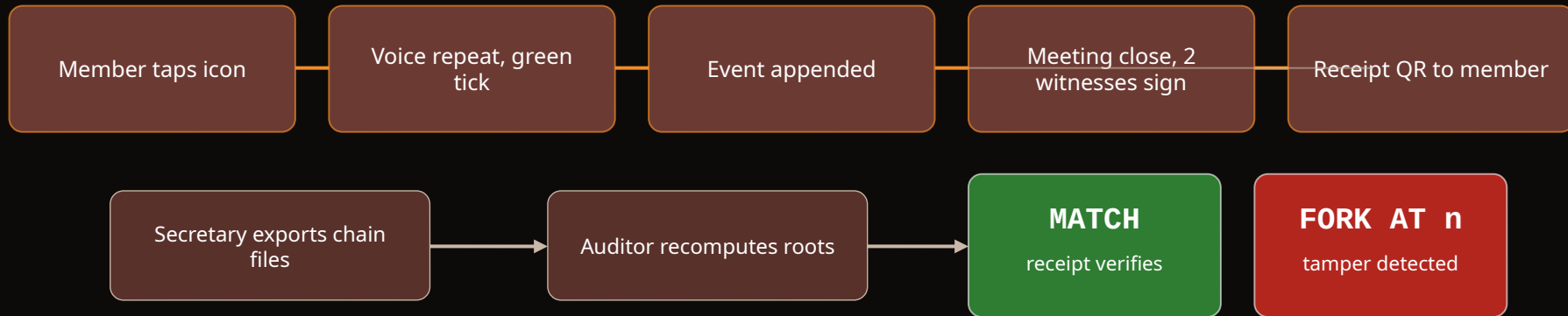
Tech Stack

- Python 3.11+ stdlib only, zero dependencies
- SHA-256 hash chain (append-only)
- HMAC-SHA256 witness MACs, demo-mode label
- SQLite transactional state
- Static HTML/CSS/JS UI, loopback server
- QR receipts, printable, byte-short
- Fully offline, zero APIs at runtime

Methodology

- 4-icon entry (contribution, loan, repayment, correction) + voice repeat
- Green-tick confirm before any event enters the chain
- Append-only: no UPDATE, no DELETE. Corrections = reversal + replacement
- 2 witness keys sign the meeting-close root
- Member receipt: root + witnesses, QR or text
- Verification recomputes every hash, offline: MATCH or FORK AT EVENT n
- Deterministic: same files, same bytes, same verdict

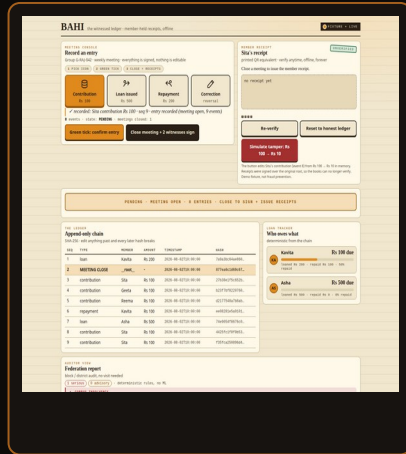
Implementation and Flow



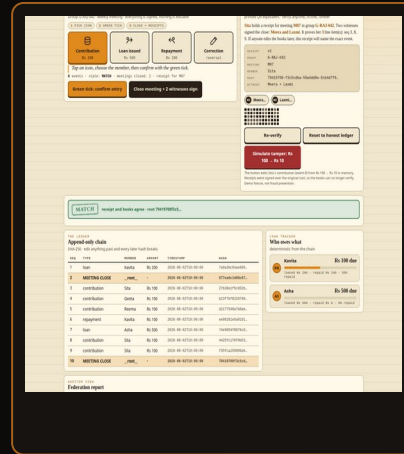
- Implementation: `chain.py`, `witness.py`, `receipt.py`, `loader.py`; attack test matrix (edit, delete, reorder, double-spend)
- Failure drill: no-network, corrupt fixture, port conflict, repeat reset. Reproducibility is auditability.

Prototype

Real demo, captured from the running app before freeze



4-icon entry + voice confirm



Member receipt (QR + root)



Fork alert: FORK*AT MEETING M07



Auditor view + export

Every screenshot is the real running app: entry, receipt, fork alert, auditor. Nothing is photoshopped.

Challenges and Risks

What we do NOT claim

- No security audit claim
- No embezzlement detection claim
- No PKI-grade signatures (witness MAC, labeled)
- No Aadhaar / OTP / KYC
- No server persistence, no live LokOS sync
- No ML. Honesty is the product
- Not fraud prevention: detection with a member-held artifact

Build risks and mitigations

- QR on feature phones: byte-short payload, text fallback
- Voice confirm on low-end audio: constrained repeat, icon fallback
- Frozen data drift: all sources saved with SHA-256 in manifest
- 7h window: stdlib only, deterministic fixtures, one-command reset
- Attack button in demo proves the detection claim live

Impact and Benefits

Member

- Owns verifiable evidence of her own money
- Receipt catches alteration instantly, offline
- No trust in bookkeeper, vendor, cloud, or chain
- Works on basic phones: print QR or copied text

Federation and system

- Block/district audit remotely, no visit
- Standardized audit report, exportable
- Verification layer over LokOS: replaces nothing, audits everything
- Tamil Nadu special audit: Rs 107.94 crore deficiencies (186 BLFs)

Scale: 10.03 crore members | 144.22 lakh savings-linked accounts | 94.16 lakh SHGs on LokOS (sources on slide 8)

Research and References

- [1] PIB PRID 2280986 (2026-07-04): LokOS 94.16 lakh SHGs, 10.03 crore members
- [2] NABARD SHG-Bank linkage stats: 144.22 lakh savings-linked accounts
- [3] sa-dhan Bharat Microfinance Report FY 2024-25: group counts and flows
- [4] v5 startup-diligence report, war1v5-ps-17: LokOS, DreamSave, Chomoka, Ensibuuko, Mifos, autopsies, M&A
- [5] FAILED-STARTUP AUTOPSIES and COMPLAINT MINING verbatim rows in repo docs/RESEARCH.md

Method: every number labeled VERIFIED or ESTIMATE with source URL + date. AI tooling used and disclosed per rulebook. Full ledger in repo docs/PROOF-LEDGER.md

What the Demo Proves

90 seconds, offline, deterministic

ENTRY

4 icons + voice repeat + green tick. No typing, no literacy barrier.

WITNESS

2 keys sign the meeting root. Receipts are QR/text, member-held.

ATTACK

Simulated tamper rewrites Rs 100 to Rs 10. Chain forks. Receipts fail.

AUDIT

Federation view lists the fork, the event, and which receipts still match.

Counter-fact: honest re-entry verifies green. The same receipt is the audit trail.

Thank You

The person who records the money is no longer the only proof.

Team 511 | Track PS-17 | github.com/harshgounder/craft-n-code-bahi

CRAFT N CODE 2.0 | ROUND 1 | 2026-08-23